A Note on Two-Stage Software Testing by Two Teams

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Abstract. This paper reports our experience of software reliability evaluation in the actual software development which employs somewhat uncommon testing procedure. The procedure can be called as a two-stage testing by two teams. First we discuss the problems underlying the software testing. We explain the concept of the procedure of the two-stage testing by two teams, and then, a stochastic model which traces the procedure is derived. The model analyzes a data set which were obtained from the real testing activity. By using the result of data analysis, we show some numerical illustrations for the software reliability assessment and the evaluation of the skill level of the test teams.

Keywords: Software reliability, Software test, Two-stage testing, Binomial distribution model.

1 Introduction

Recall news reports on various electrical/electronic industrial products have been often made in the recent mass media. Almost all of the recall news reports alert severe problems which may affect our social life, if these problems could be actualized. For example, such problems have actually occurred in some braking equipment or the steering control system in automobiles, and in the other cases, somebody has experienced the problem in which he/she could not dial the emergency call of three figures in the mobile phone. One of the most conceivable causes is the lack of reliability in the program code included in these software systems and products.

In such a situation, in the testing activities in the software development process or the simulated operational testing, software developers verify the software which is to be implemented in terms of the conformity with the customer's requirements, by executing huge amount and various test cases in order to check whether the software behaves and performs without any problem.

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Today, the software developers test the software so as to find all of the malfunctions which may occur in the operational environment, by using the traceability matrix [1] made of the information on which test cases are tested and how those verify the customer's requirements.

Even though we release such precisely-tested software systems, we have often experienced the following phenomena:

- Some faults are still detected in the customer's acceptance testing.
- Some claims of the product are sent from the customers after purchasing in a mass market.

Although the software developers apply the traceability matrix to the software not to overlook the customer's requirements and sufficiently test it, why do the above-mentioned problems occur?

In order to solve these difficult problems, various testing methods have been studied and proposed by many researchers, real developers, and practitioners so far. These are for instance, capture-recapture method [2,3], two-stage editing method [4], and so on [1,5]. Also in recent years, Akiyama et al. have proposed the test case design method which is applied the Latin square approach (see e.g. [6]).

Recently we have obtained a development opportunity of a certain application software which runs on both of Windows7 OS of 32 bit and 64 bit versions. On the development, first of all we have carried out all tests on the 32 bit OS. After that, we have tested it again on the 64 bit OS, as a regression test. In these testing, we used the same test cases to this identical software program (i.e., object code). We first expected that any fault would not be detected in the regression testing, however, in fact, several faults have been detected.

Therefore we have investigated the causes of these faults detection in the regression testing. We thought such causes would surely lead to the software failure occurrence in the acceptance testing by the customer and in the users environment after the release. As the result, we have been able to divide the causes into two groups as follows:

- Lack of the testers' understanding of the contents of the assigned test cases.
- Lack of confirmation to the each result of the test by the testers.

About the first cause pointed above, the overlooking probability of a fault can be reduced by reflecting in the review check lists to the design specifications and the testing specifications. On the other hand, for the second point above, it becomes avoidable by introducing the automatic testing or applying the testing tool.

Hence in this research, we propose the reliability prediction method for assessing the testing activities based on the data which were collected in such testing environment. Especially, we show the prediction of the number of remaining faults and the evaluation of the testing skill, in order to clarify the appropriate testing activities which should be performed and estimate the number of faults which should be detected in the acceptance test phase.